

Raw coupled prediction

$$c_{raw} = \hat{R}^{-1}d_*$$

Both checks
pass?

Yes

Set $\hat{c} = c_{raw}$

No

**Project for mass
conservation**

Compute c_{aff}

Both checks
pass?

Yes

Set $\hat{c} = c_{aff}$

No

Weighted projection

Minimize $w_c^\top \rho$

$$Ac = Ac_{in,*}, \quad c \geq 0$$

$$\rho \geq |c - c_{aff}|$$

Solve and
audit pass?

Yes

Use optimized $\hat{c}$

No

Prediction failure

Admissibility not verified

Checked output

$\hat{c}, \quad I_{\text{comp}}\hat{c}$